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GADJAH MADA

Linear Classifier – Single Layer Perceptron

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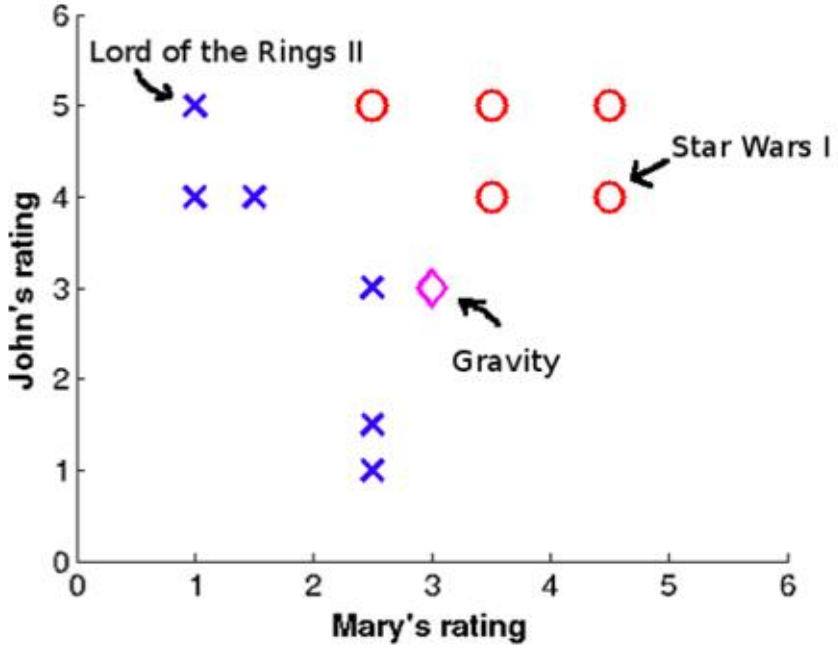
Linear Classifier

Movie name	Mary's rating	John's rating	I like?
Lord of the Rings II	1	5	No
...	...	...	...
Star Wars I	4.5	4	Yes
Gravity	3	3	?



Linear Classifier

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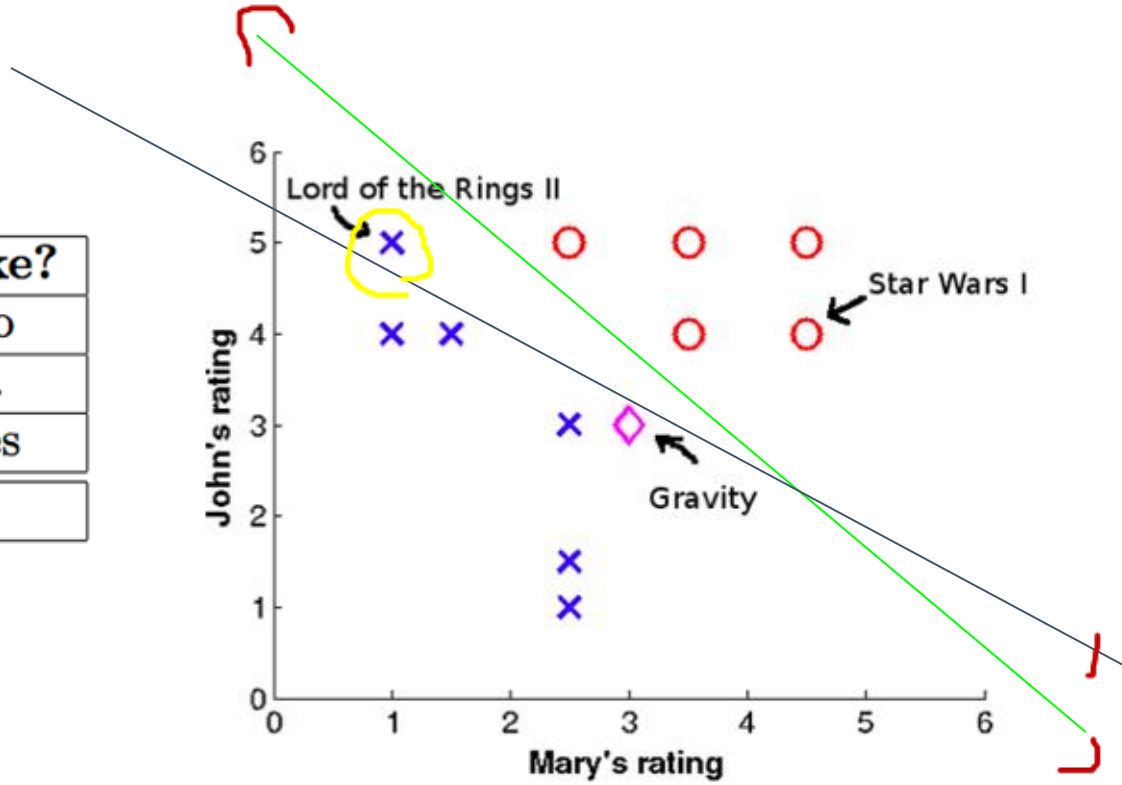
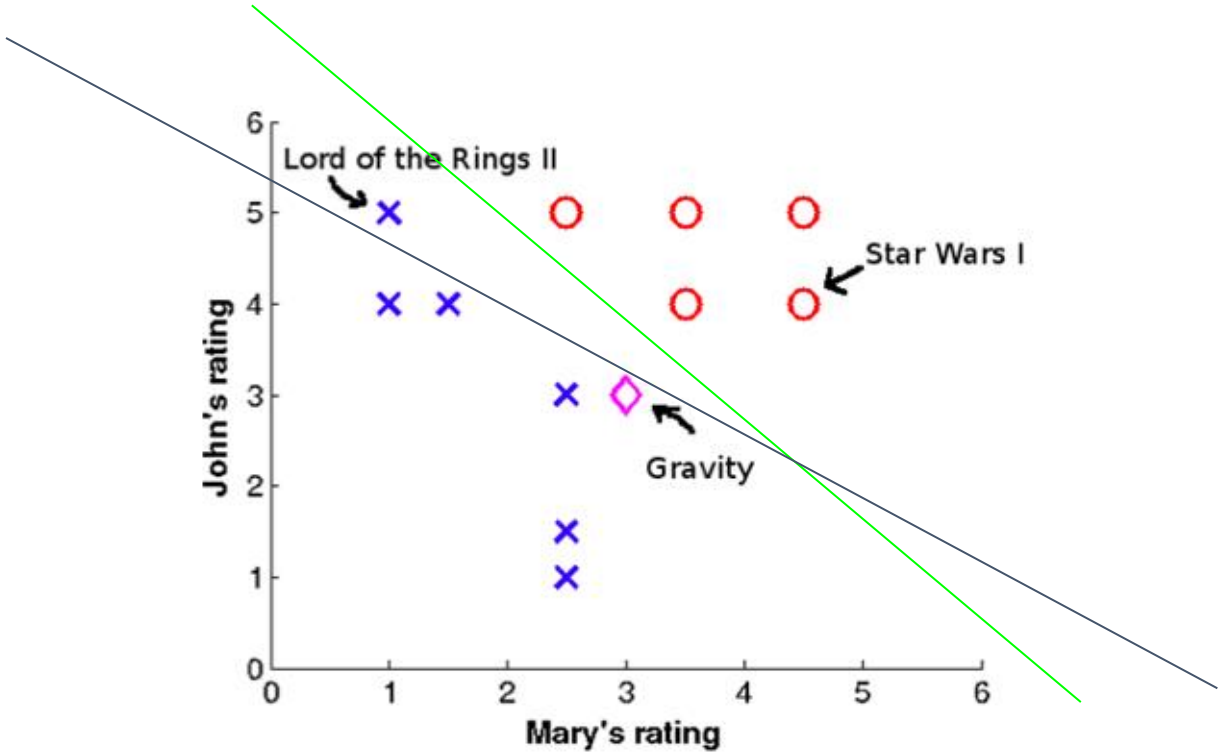


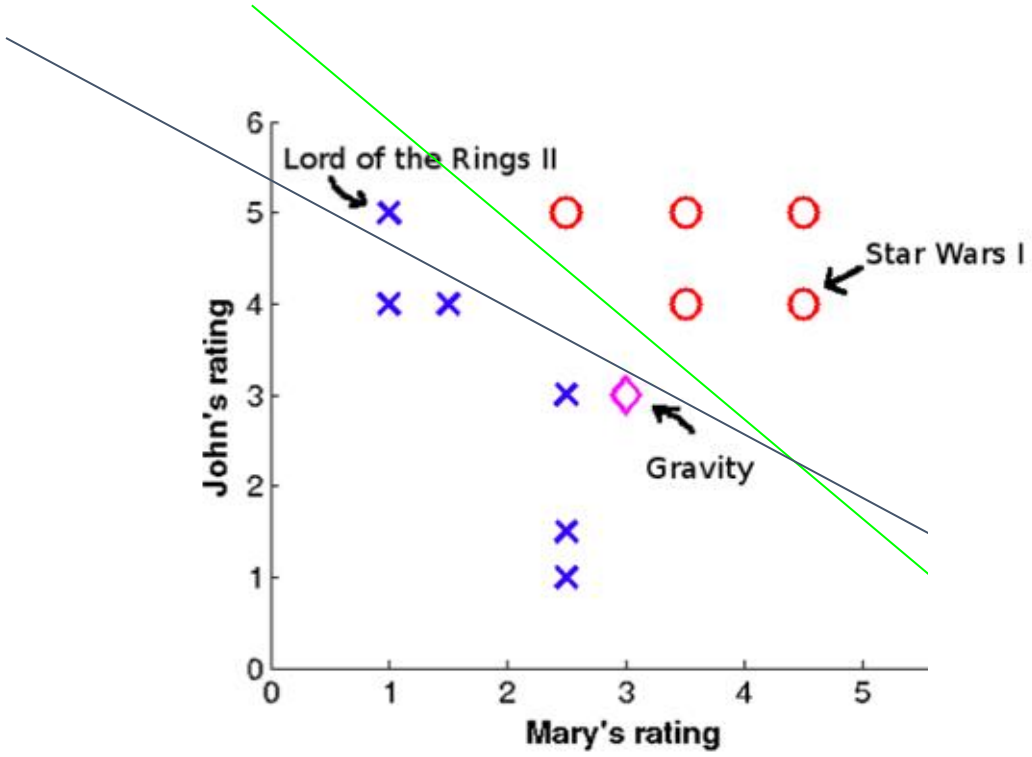
Linear Classifier



$$h(x;\theta,b) = \theta_1 x_1 + \theta_2 x_2 + b$$

$$h(x;\theta,b) = \theta^T x + b$$

Linear Classifier



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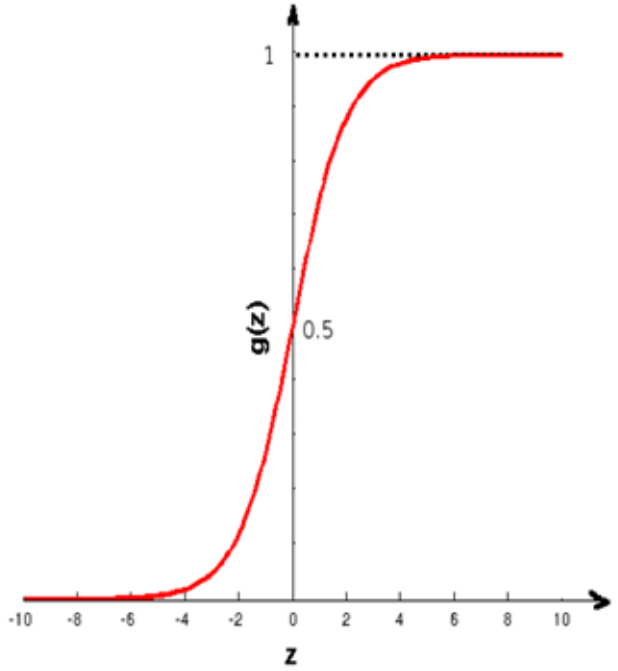
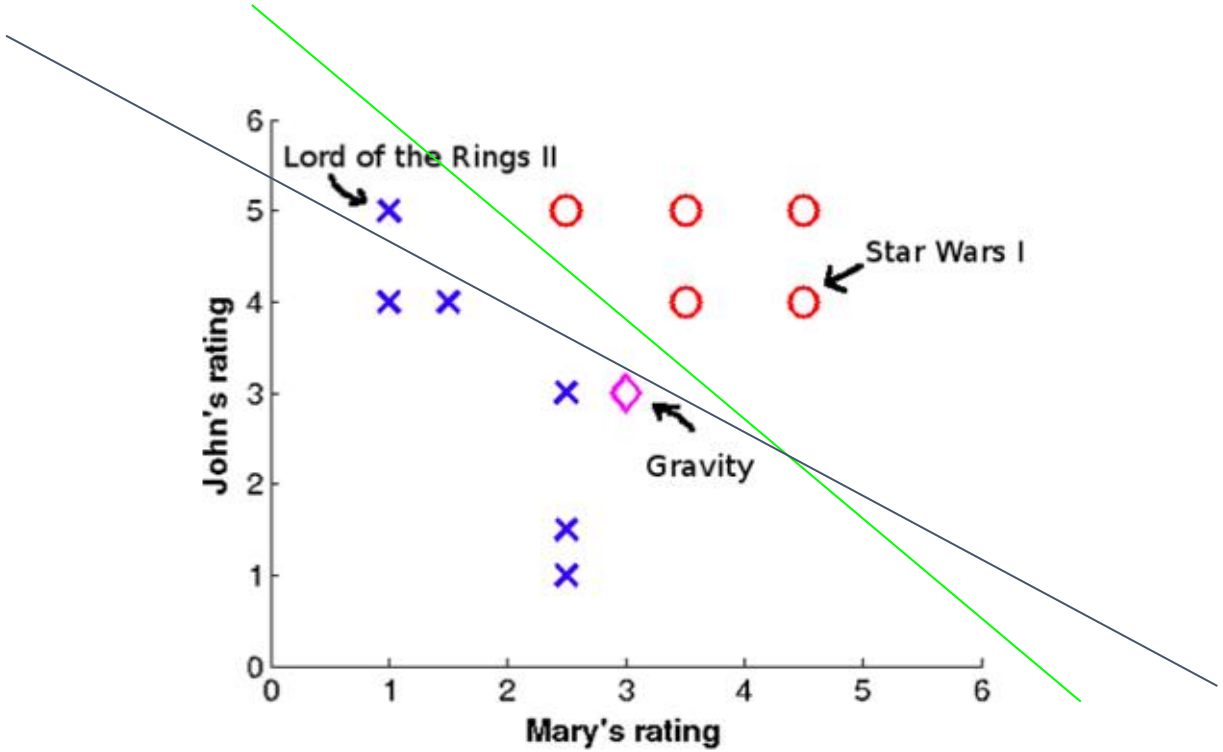
$$h(x;\theta,b) = \theta^T x + b$$



$$h(x;\theta,b) = g(\theta^T x + b)$$

$$g(z) = \frac{1}{1+\exp(-z)}$$

Linear Classifier – Single Layer Perceptron (SLP)

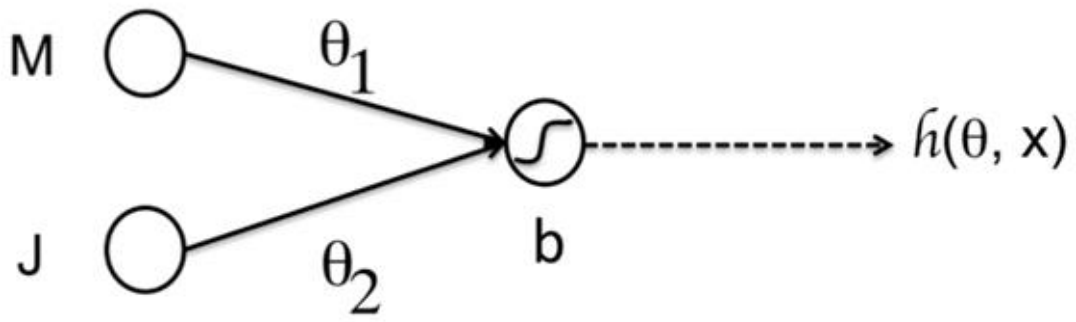


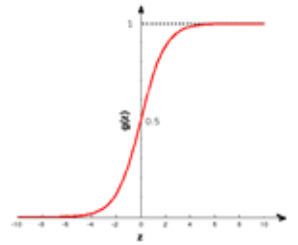
$$h(x;\theta,b) = \theta_1 x_1 + \theta_2 x_2 + b$$

$$h(x;\theta,b) = \theta^T x + b$$

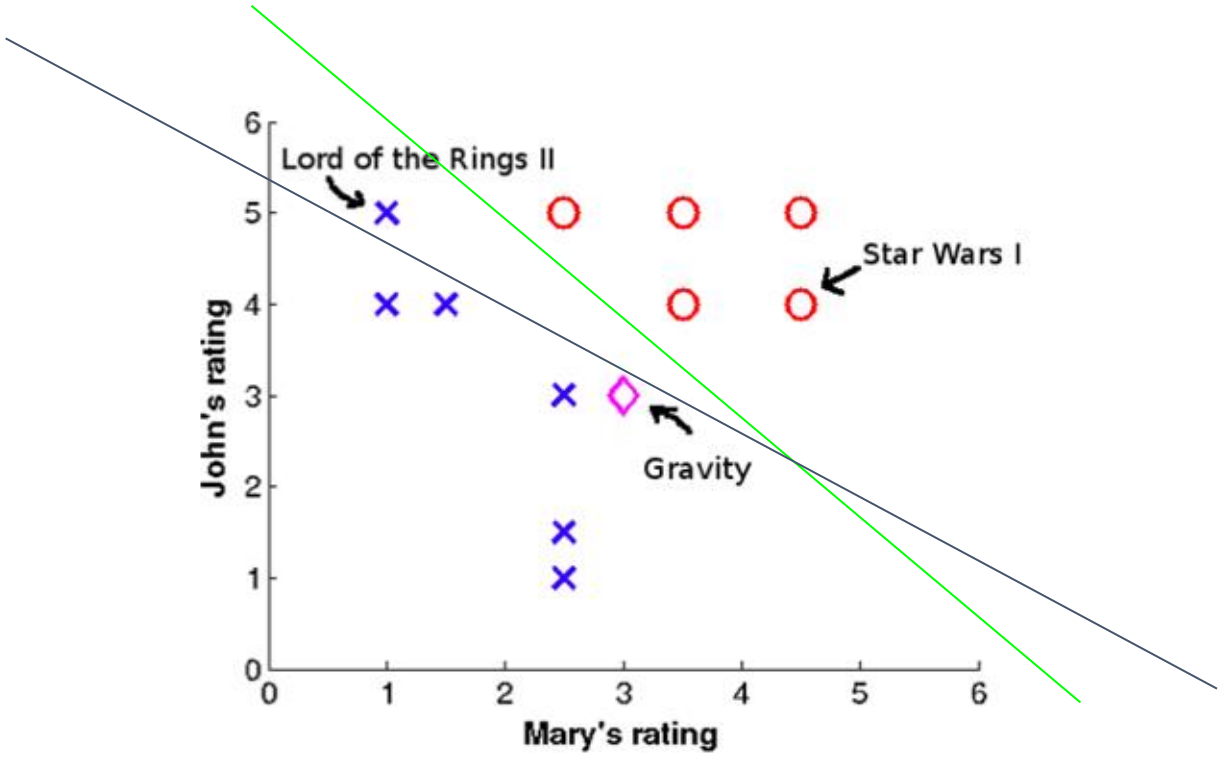
$$h(x;\theta,b) = g(\theta^T x + b)$$

$$g(z) = \frac{1}{1+\exp(-z)}$$





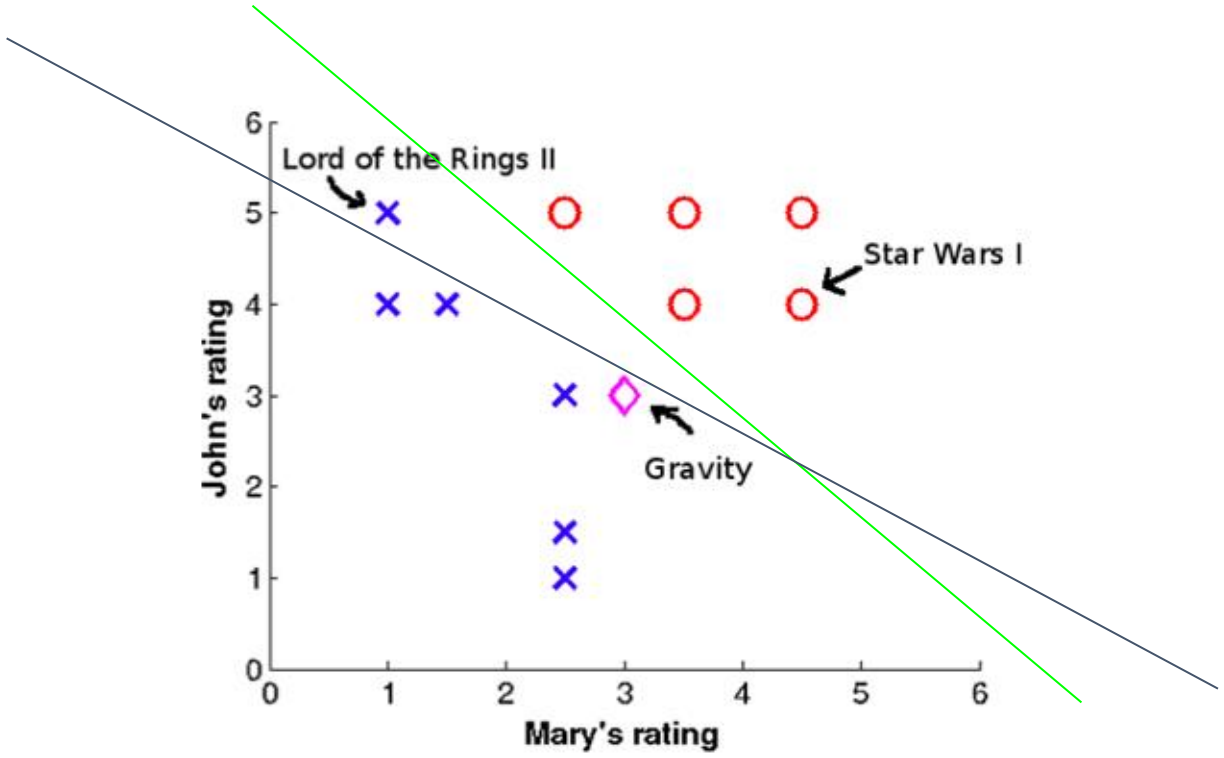
Linear Classifier – Single Layer Perceptron (SLP)



How to get appropriate linear classifier ?

LEARNING from Error

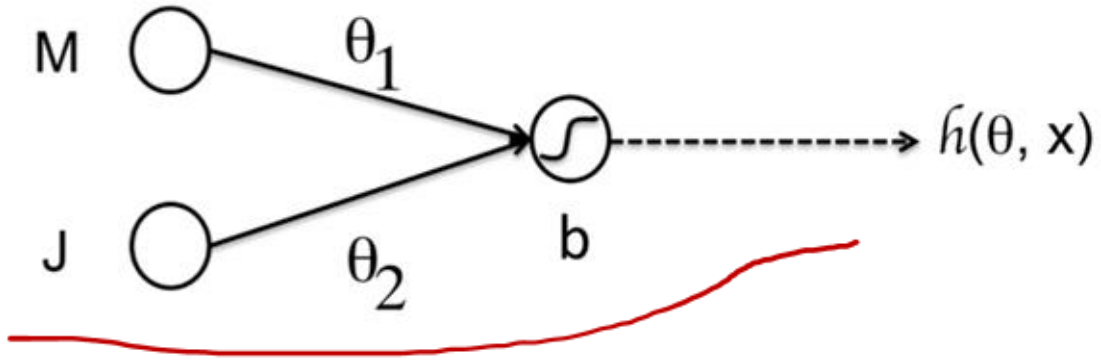
Single Layer Perceptron (SLP) – Stochastic Gradient Descent



LEARNING
from Error

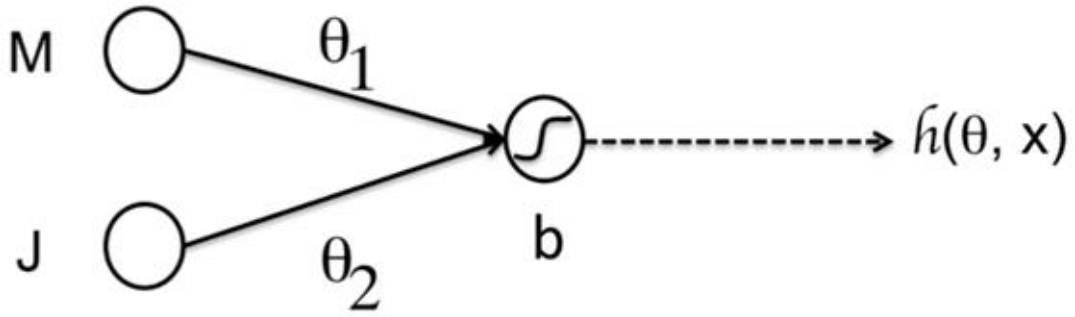
Error function
Cost function
Lost Function

Single Layer Perceptron (SLP) – Gradient Descent



$$\theta_1 = \theta_1 - \alpha\Delta\theta_1$$
$$\theta_2 = \theta_2 - \alpha\Delta\theta_2$$
$$b = b - \alpha\Delta b$$

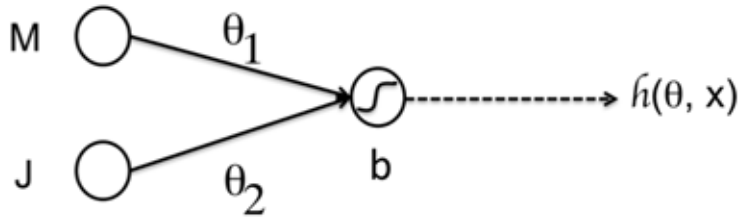
Single Layer Perceptron (SLP) – Gradient Descent



$$\theta_1 = \theta_1 - \alpha\Delta\theta_1$$
$$\theta_2 = \theta_2 - \alpha\Delta\theta_2$$
$$b = b - \alpha\Delta b$$

$$\begin{aligned}\Delta\theta_1 &= \frac{\partial}{\partial\theta_1}\left(h(x^{(i)};\theta,b) - y^{(i)}\right)^2 \\ &= 2\left(h(x^{(i)};\theta,b) - y^{(i)}\right)\frac{\partial}{\partial\theta_1}h(x^{(i)};\theta,b) \\ &= 2\left(g(\theta^T x^{(i)} + b) - y^{(i)}\right)\frac{\partial}{\partial\theta_1}g(\theta^T x^{(i)} + b)\end{aligned}$$

Single Layer Perceptron (SLP) – Gradient Descent



$$\Delta\theta_1 = \frac{\partial}{\partial\theta_1}\left(h(x^{(i)};\theta,b) - y^{(i)}\right)^2$$
$$= 2\left(h(x^{(i)};\theta,b) - y^{(i)}\right)\frac{\partial}{\partial\theta_1}h(x^{(i)};\theta,b)$$
$$= 2\left(g(\theta^T x^{(i)} + b) - y^{(i)}\right)\frac{\partial}{\partial\theta_1}g(\theta^T x^{(i)} + b)$$

$$\frac{\partial g}{\partial z} = [1 - g(z)]g(z)$$

$$\frac{\partial}{\partial\theta_1}g(\theta^T x^{(i)} + b) = \frac{\partial g(\theta^T x^{(i)} + b)}{\partial(\theta^T x^{(i)} + b)}\frac{\partial(\theta^T x^{(i)} + b)}{\partial\theta_1}$$
$$= [1 - g(\theta^T x^{(i)} + b)]g(\theta^T x^{(i)} + b)\frac{\partial(\theta_1 x_1^{(i)} + \theta_2 x_2^{(i)} + b)}{\partial\theta_1}$$
$$= [1 - g(\theta^T x^{(i)} + b)]g(\theta^T x^{(i)} + b)x_1^{(i)}$$

Single Layer Perceptron (SLP) - Gradient Descent

$$\begin{aligned}\Delta\theta_1 &= \frac{\partial}{\partial\theta_1}\left(h(x^{(i)};\theta,b)-y^{(i)}\right)^2 \\ &= 2\left(h(x^{(i)};\theta,b)-y^{(i)}\right)\frac{\partial}{\partial\theta_1}h(x^{(i)};\theta,b) \\ &= 2\left(g(\theta^T x^{(i)}+b)-y^{(i)}\right)\frac{\partial}{\partial\theta_1}g(\theta^T x^{(i)}+b)\end{aligned}$$

$$\begin{aligned}\Delta\theta_1 &= 2\left[g(\theta^T x^{(i)}+b)-y^{(i)}\right]\left[1-g(\theta^T x^{(i)}+b)\right]g(\theta^T x^{(i)}+b)x_1^{(i)} \\ \Delta\theta_2 &= 2\left[g(\theta^T x^{(i)}+b)-y^{(i)}\right]\left[1-g(\theta^T x^{(i)}+b)\right]g(\theta^T x^{(i)}+b)x_2^{(i)} \\ \Delta b &= 2\left[g(\theta^T x^{(i)}+b)-y^{(i)}\right]\left[1-g(\theta^T x^{(i)}+b)\right]g(\theta^T x^{(i)}+b)\end{aligned}$$

$$\begin{aligned}\frac{\partial}{\partial\theta_1}g(\theta^T x^{(i)}+b) &= \frac{\partial g(\theta^T x^{(i)}+b)}{\partial(\theta^T x^{(i)}+b)}\frac{\partial(\theta^T x^{(i)}+b)}{\partial\theta_1} \\ &= \left[1-g(\theta^T x^{(i)}+b)\right]g(\theta^T x^{(i)}+b)\frac{\partial(\theta_1 x_1^{(i)}+\theta_2 x_2^{(i)}+b)}{\partial\theta_1} \\ &= \left[1-g(\theta^T x^{(i)}+b)\right]g(\theta^T x^{(i)}+b)x_1^{(i)}\end{aligned}$$

Single Layer Perceptron (SLP) – Gradient Descent

$$\Delta\theta_1 = 2\left[g(\theta^T x^{(i)} + b) - y^{(i)}\right]\left[1 - g(\theta^T x^{(i)} + b)\right] g(\theta^T x^{(i)} + b) x_1^{(i)}$$
$$\Delta\theta_2 = 2\left[g(\theta^T x^{(i)} + b) - y^{(i)}\right]\left[1 - g(\theta^T x^{(i)} + b)\right] g(\theta^T x^{(i)} + b) x_2^{(i)}$$
$$\Delta b = 2\left[g(\theta^T x^{(i)} + b) - y^{(i)}\right]\left[1 - g(\theta^T x^{(i)} + b)\right] g(\theta^T x^{(i)} + b)$$

1. Initialise the weight and bias randomly
2. Calculate the error function, if error still big then do
   a. Update the weight
   b. Update the bias

Hands on – Iris Dataset

- https://www.kaggle.com/datasets/uciml/iris
- 3 class @50 => total : 150 data
- #feature = 4 features
- Training = 120
- Testing = 30 (10 data terakhir dari setiap kelasnya)
- Define an architecture of SLP

Hands on – Iris Dataset - spreadsheet

- https://www.kaggle.com/datasets/uciml/iris
- 2 class @50 => total : 100 data
  - Iris Setosa
  - Iris Versicolour
- #feature = 4 features
- Training = 80
- Testing = 20
- Define and architecture of SLP
  - Output neuron => binary encoding

Hands on – Iris Dataset - spreadsheet

- Spreadsheet https://docs.google.com/spreadsheets/d/1yhi8ASebpmljJ3NpYQ7NmxlWLmhGOI6W/edit?usp=sharing&ouid=117122342833524350176&rtpof=true&sd=true